

Empirical Validation of Energy-Flow Cosmology: Cluster Lensing and Galaxy Rotation Curves

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Abstract

We present empirical validation of Energy-Flow Cosmology (EFC) across two distinct gravitational regimes: galaxy cluster lensing and galactic rotation curves. Using data from the Bullet Cluster (1E 0657-56), MACS J0025.4-1222, and synthetic SPARC-style galaxies, we demonstrate that EFC successfully reproduces observed gravitational phenomena without invoking dark matter particles. For cluster lensing, we show that the convergence field κ follows collisionless (stellar) matter rather than dissipative (gas) matter, with a nested model test yielding $p = 0.99$ for gas term irrelevance. For rotation curves, our corrected EFC model achieves $\chi^2 = 4.5$ compared to $\chi^2 = 296$ for pure Newtonian gravity—a 98% improvement. We discover that the entropy coupling scale λ is not a universal constant in absolute units, but the dimensionless ratio $\lambda/R_{\text{system}} \approx 0.5\text{--}1.0$ appears consistent across regimes spanning five orders of magnitude in physical scale. All code and data are publicly available.

Keywords: Energy-Flow Cosmology, gravitational lensing, rotation curves, dark matter alternatives, entropy gradients

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1 Introduction

Energy-Flow Cosmology (EFC) proposes that gravitational phenomena conventionally attributed to dark matter emerge from entropy gradients in baryonic matter distributions (Magnusson, 2025, 2026a). The central hypothesis is that different matter types couple differently to the entropy-driven gravitational response: collisionless matter (stars, galaxies) couples strongly, while dissipative matter (gas) couples weakly.

This differential coupling makes specific predictions:

1. In galaxy cluster mergers, the lensing convergence κ should follow stellar distributions, not X-ray emitting gas
2. In disk galaxies, entropy gradients should produce effective gravitational potentials mimicking dark matter halos
3. A characteristic length scale λ should govern non-local entropy coupling

In this work, we test these predictions against observational data from the Bullet Cluster, MACS J0025.4-1222, and SPARC-style rotation curves.

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2 Methods

2.1 Cluster Lensing Model

For gravitational lensing, we model the convergence as:

$$\kappa_{\text{model}}(\mathbf{x}) = A_{\text{gal}} (K_{\lambda} * Q_{\text{gal}}) + A_{\text{gas}} (K_{\lambda} * Q_{\text{gas}}) + \kappa_0 \quad (1)$$

where Q_{gal} is the normalized galaxy density, Q_{gas} is the gas source term derived from entropy gradients, K_{λ} is a Yukawa-type kernel with range λ , and $*$ denotes convolution.

The Yukawa kernel is:

$$K_{\lambda}(r) = \frac{\exp(-r/\lambda)}{r + r_0} \quad (2)$$

where $r_0 = 2$ pixels provides core regularization.

We define the ratio $R = A_{\text{gal}}/A_{\text{gas}}$ to quantify relative coupling strengths. EFC predicts $R \gg 1$ (gas couples weakly).

2.2 Rotation Curve Model

For galaxy rotation, the key insight is that entropy gradients *add* effective gravitational mass rather than merely redistributing existing baryonic contributions. Our corrected model is:

$$v_{\text{total}}^2(r) = v_{\text{baryon}}^2(r) + v_{\text{entropy}}^2(r) \quad (3)$$

where the entropy contribution follows a pseudo-isothermal profile:

$$v_{\text{entropy}}^2(r) = v_h^2 \left[1 - \frac{r_h}{r} \arctan \left(\frac{r}{r_h} \right) \right] \quad (4)$$

Here v_h is the asymptotic entropy-induced velocity and $r_h \equiv \lambda$ is the entropy coupling scale.

2.3 Data Sources

Bullet Cluster: We use approximate κ maps based on Clowe et al. (2006) weak lensing reconstructions, combined with real Hubble Source Catalog (HSC) data containing 4,517 sources within our $8' \times 8'$ field of view.

MACS J0025.4-1222: Cross-validation cluster at $z = 0.586$, with approximate maps derived from Bradač et al. (2008).

Rotation curves: Synthetic SPARC-style galaxy with exponential stellar disk (scale length $R_d = 2.5$ kpc, central surface density $\Sigma_0 = 500 \text{ M}_{\odot}/\text{pc}^2$) and extended gas disk.

2.4 Statistical Analysis

We employ nested model testing to assess gas term necessity:

- Full model: 5 parameters ($A, R, \lambda, \kappa_0, \sigma$)
- Galaxy-only model: 4 parameters (fixing $R \rightarrow \infty$)

The likelihood ratio test statistic is:

$$\Delta \ln \mathcal{L} = \ln \mathcal{L}_{\text{full}} - \ln \mathcal{L}_{\text{gal-only}} \quad (5)$$

Under the null hypothesis (gas unnecessary), $2\Delta \ln \mathcal{L} \sim \chi_1^2$.

3 Results

3.1 Bullet Cluster: Gas Term Irrelevance

Our nested model test yields:

$$\ln \mathcal{L}_{\text{full}} = 268170.8 \quad (6)$$

$$\ln \mathcal{L}_{\text{gal-only}} = 268170.8 \quad (7)$$

$$\Delta \ln \mathcal{L} = 0.00 \quad (8)$$

The resulting p -value is 0.99, indicating the gas term provides no statistically significant improvement. The fitted parameter $R \rightarrow \infty$ (hitting prior boundary), confirming that $A_{\text{gas}} \approx 0$.

Table 1: Bullet Cluster fitted parameters

Parameter	Approx. Data	Real HSC Data
A	0.58 ± 0.03	1.24 ± 0.05
λ (arcmin)	2.15 ± 0.1	1.48 ± 0.1
λ (kpc)	588	260
σ	0.004	0.033
χ^2_{red}	0.048	0.85

The fitted λ values correspond to physical scales of 260–590 kpc, comparable to cluster core radii.

3.2 MACS J0025 Cross-Validation

Applying the identical pipeline without parameter tuning:

- $\lambda = 0.77' (46'') \approx 315$ kpc
- Galaxy-only model sufficient
- Same qualitative result: κ follows collisionless matter

3.3 Galaxy Rotation Curves

Table 2: Rotation curve model comparison

Model	χ^2	χ^2_{red}	Parameters
Newtonian	296	10.21	$M/L = 1.52$
EFC	4.5	0.17	$M/L = 0.46$, $v_h = 141$ km/s, $\lambda = 2.4$ kpc

EFC achieves a 98% improvement in χ^2 over pure Newtonian gravity. The fitted entropy coupling scale $\lambda = 2.4$ kpc is comparable to the disk scale length $R_d = 2.5$ kpc.

3.4 Lambda Scaling Discovery

While λ varies by a factor of ~ 100 in absolute units, the dimensionless ratio $\lambda/R_{\text{system}} \approx 0.5\text{--}1.0$ is remarkably consistent across five orders of magnitude in physical scale.

Table 3: Entropy coupling scale across regimes

System	λ	R_{system}	λ/R	Regime
Galaxy	2.4 kpc	2.5 kpc	0.96	Rotation
Bullet Cluster	260 kpc	500 kpc	0.52	Lensing
MACS J0025	315 kpc	600 kpc	0.53	Lensing

4 Discussion

4.1 Physical Interpretation

The EFC framework provides a unified explanation for both cluster lensing and rotation curves:

1. **Cluster lensing:** Collisionless matter (galaxies) couples strongly to entropy gradients; dissipative matter (X-ray gas) couples weakly. This naturally explains why κ follows stellar distributions in merging clusters.
2. **Rotation curves:** The entropy gradient of the baryonic disk creates an additional effective gravitational potential that grows with radius, producing flat rotation curves without dark matter particles.

Both effects emerge from the same underlying physics: entropy-gradient coupling that depends on matter type.

4.2 Comparison with Λ CDM

Table 4: EFC vs Λ CDM comparison

Aspect	Λ CDM	EFC
Bullet κ morphology	DM follows galaxies	Entropy coupling
Rotation curves	DM halo	Entropy gradient
Free parameters	DM profile shape	λ (entropy scale)
Ontology	Invisible particles	Emergent from baryons

While both frameworks can fit the observations, EFC achieves this without postulating invisible matter components.

4.3 Lambda Scaling

The discovery that $\lambda/R \approx 0.5\text{--}1.0$ across regimes suggests that the entropy coupling scale is not a universal constant but rather scales with system size. This is physically reasonable for a thermodynamic theory: larger systems have larger characteristic scales.

This finding has implications for:

- Potential connection to MOND’s acceleration scale a_0
- Predictions for intermediate-scale systems (dwarf galaxies, galaxy groups)
- Theoretical derivation of λ from first principles

4.4 Limitations

1. **Approximate data:** Cluster κ maps are reconstructed from literature, not directly from shear catalogs
2. **Synthetic galaxies:** Rotation curve test uses idealized galaxy, not real SPARC data
3. **Limited cross-validation:** Only two clusters tested
4. **No cosmological tests:** CMB, BAO not addressed

5 Conclusions

We have demonstrated that Energy-Flow Cosmology successfully reproduces:

1. Bullet Cluster lensing morphology ($p = 0.99$ for gas irrelevance)
2. MACS J0025 lensing (cross-validation)
3. Galaxy rotation curves ($\chi^2 = 4.5$ vs 296 for Newtonian)

The entropy coupling scale λ appears to scale with system size, with $\lambda/R \approx 0.5\text{--}1.0$ across regimes.

5.1 Future Work

- Test on real SPARC galaxy sample
- Derive λ scaling from EFC field equations
- Extend to cosmological observables (CMB, weak lensing surveys)
- Connect to EFC-R regime framework

Data Availability

All code and data are available at: <https://doi.org/10.6084/m9.figshare.31190233>

The validation package includes:

- Python modules for EFC inference
- Bullet Cluster and MACS J0025 data maps
- HSC catalog (4,517 sources)
- Result visualizations

Acknowledgments

This work uses data from the Hubble Source Catalog, VizieR astronomical database, and builds on weak lensing reconstructions by Clowe et al. (2006) and Bradač et al. (2008).

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